

Certificate of correctness for the Lean 4 formalization accompanying *Uniform rationality of p -adic factorization densities*

Companion to the manuscript

Lean 4 v4.31.0 / mathlib v4.31.0
Ledger re-verified 2026-06-22

Abstract

This is the “more certainty than usual” companion to the manuscript and the released Lean code in `lean.urat/`. We are exact about what the formalization establishes. The capstone theorems are genuine Lean **theorems**; their machine-checked `#print axioms` footprints are recorded here verbatim, regenerated by the audit harness `LeanUrat/AxChk_baseline.lean`. The development is *verified modulo an explicit list* of assumptions carried as structure fields, theorem hypotheses, and one named literature-cited axiom: **(A1)** the minimal box-wise p -adic Haar / equidistribution identification, **(A2)** the Guàrdia–Montes–Nart (GMN) count-shadow hypotheses, and **(A3)** the tame functional equation. It is **not** an unconditional machine-checked proof: `mathlib v4.31.0` has no p -adic Haar measure, so the density itself cannot be constructed in Lean — the only inhabited instance is a trivial degree-0 witness. We state this limitation — the *measure wall* — plainly. The accompanying symbolic and finite-field computations are presented as *evidence*, never as part of the formal proof; they are runnable (Section 7).

Contents

1	Purpose: what this certificate is, and what it is not	2
2	The assumed ledger (the explicit <i>modulo</i> list)	2
3	The capstone ledger (<code>#print axioms</code>, verbatim)	3
3.1	The two Goal theorems (the headline)	4
3.2	Rationality / decomposition (count-native, only-Montes) — all CORE	4
3.3	The proved building blocks — all CORE	4
3.4	The measure layer (<code>PadicMeasure</code> , <code>L6M4</code> , <code>BB3inf</code>) — where the wall lives	5
4	Which parts of the paper’s argument are formalized	6
5	The measure wall (the honest limitation)	7
6	Pre-existing flagged sorrys (off every capstone footprint)	7
7	Independent cross-checks (EVIDENCE, not proof)	7
7.1	Closed-form arithmetic the Lean modules prove (exact)	8
7.2	The symbolic OM engine reproduces BCFG root-count moments through $n = 6$. . .	8
7.3	$n \leq 3$ exact closed forms vs BCFG, and the wild atoms	9
8	Release module list	10

1 Purpose: what this certificate is, and what it is not

What it is. A faithful record of a *machine-checked-modulo-explicit-axioms* formalization plus a set of *independent cross-checks*. Concretely:

- The paper’s two main statements — *uniform rationality* and *palindromy* of the projective p -adic factorization-type density — are reduced in Lean to an explicit minimal interface. The conclusions (rationality, palindromy, the tree decomposition $\rho = \sum_T C_T$) are Lean **theorems**, never axioms. Everything available in `mathlib` is used from `mathlib`; everything cited from the literature is a named **axiom** or a hypothesis field with a doc-comment citation.
- The whole development is *conditional* on the assumed ledger (A1)–(A3) of Section 2, carried as hypotheses/axioms.
- Independently, a symbolic Okutsu–Montes (OM) density engine reproduces the Bhargava–Cremona–Fisher–Gajović (BCFG) root-count distribution and moments *exactly* through $n = 6$, and a p -adic factorization oracle confirms the wild atoms numerically. These are *evidence* for the recursion, not part of the proof.

What it is NOT. It is **not** an unconditional proof. The honest one-line summary:

Verified in Lean 4 modulo $\{(A1) \text{ the minimal box-wise } p\text{-adic Haar identification (the measure wall); (A2) \text{ the GMN count-shadow hypotheses; (A3) \text{ the tame functional equation}\}$, plus Lean’s three core classical axioms.

Never “formally verified” bare. The paper’s density $\rho(n, \sigma; q)$ cannot be constructed in Lean (Section 5); the conjectural r -point / pole-control refinements are **not** claimed as proven anywhere here.

How to read the ledger. For each declaration we list its exact `#print axioms` footprint. We write `CORE` for `{propext, Classical.choice, Quot.sound}` — Lean’s core classical foundations, equivalent to ordinary classical mathematics, present in essentially every classical theorem; we do not count them as project assumptions. Any *additional* name is a project axiom and is part of the explicit *modulo* list for that declaration. There is **no** `sorryAx` in any of the 55 footprints audited (Section 3).

2 The assumed ledger (the explicit *modulo* list)

The development is conditional on exactly the following, beyond `CORE`. None asserts rationality, uniformity, or palindromy; each is a structural measure/combinatorial fact absent from `mathlib v4.31.0`, recorded in the established “recorded-via-hypothesis” style.

(A1) The minimal box-wise measure axiom — the single irreducible measure residue. Carried as **two structure fields** of `MontesAxiom.MontesData` (not a global **axiom**; hypothesis-discharged, so it does *not* surface in `#print axioms`):

- `boxHaarEquidist`: per-box Haar volume equals the proved closed value,

$$\text{boxMeasure } c \ q' = \text{countCellCoeff } c \ q' = (q^\delta)^{d_S-1} \cdot \text{bb1Value } c.\text{polygon } q'.$$

That is: the abstract per-box p -adic Haar volume **equals** the separately *proved* residual-count $\times$ box-volume value.

- `nodeMeasure_boxSum`: the node limit *exists* and is *box-additive* — $\lim_N \text{stratumCount } T \, N / q^{nN}$ equals the box-additive assembly $\sum_{\text{cells } c} \text{boxMeasure } c \, q \cdot \prod_{\text{children}} \text{clusterCount} / \text{countPivot}$.

The right-hand sides (the `bb1Value` box volume, the $(q^\delta)^{ds-1}$ residual count, the box-additivity arithmetic) are all *proved* (P1–P3 below); only the *identification* of the abstract Haar volume with that proved value is assumed. This is the genuine **measure wall** (Section 5).

(A2) The GMN count-shadow hypotheses. The remaining `MontesData` / `CountingModel` structure fields, none of which carries the conclusion: `shapesOf`, `partition` (unique-tree disjoint partition `decidedCount` $\sigma \, N = \sum_T \text{stratumCount } T \, N$); `treeSize`, `cells`, `cells_descend` (strict cluster-size drop on descent — the count shadow of `omCells/descend_size_lt`, supplying well-foundedness); `finiteTermination` (GMN Thm 4.18, $\leq n$ leaves); and the elementary analytic-tail bounds `undecided_le_split`, `sepTail_tendsto`, `discriminantTail_envelope` (bounds, *not* the conclusion $U_N \rightarrow 0$, which is *derived*; see P6). There is **no** “ $U_N \rightarrow 0$ ” field and **no** “ C_T is rational” field — those are theorems.

(A3) tame_functionalEquation (the one declared axiom, `Interface.lean`). The genuine §5 H-tame input: an abstract `DensityFoundation`’s density admits a palindromic rational representative at tame primes (Del Corso–Dvornicich 2000; paper §5). This is the **only** non-core declared axiom on a capstone footprint; it appears *only* on `goal_theorem_montes` (and the witness instance), never on any pure-rationality constant. It asserts nothing about the count-native $\sum_T C_T$ directly — the link is the hypothesis `hbridge`.

(A4) hbridge (hypothesis argument, not an axiom). $\forall q' > 1, F.\text{density } n \, \sigma \, q' = \sum_T \text{coeff } T \, q'$. The honest measure-wall identification (“abstract foundation density coincides with the count-native OM tree-sum”); RHS is an arbitrary function, asserting nothing about a closed form. Introduces no axiom.

The disconnected measure-route axioms (NOT on the capstone path). The module `LeanUrat.PadicMeasure` declares genuine axioms — `clusterMeasure`, `AX_cellRecursion`, `AX_columnMeasure`, `omCells`, `descend`, `descend_size_lt` (and `BB3inf` declares its own parallel `BB3inf.descend`, `BB3inf.descend_size`). These are the *older measure route*. They are verified **absent** from `goal_theorem_montes` and from every count-native rationality/decomposition constant (Section 3); they survive only on the historical `goal_theorem` and the `PadicMeasure/L6M4` declarations, kept for the original derivation. Their content: `AX_cellRecursion` (Igusa–Denef per-cell decomposition; Igusa *Local Zeta Functions* §7.4, Denef Invent. Math. 77 (1984)); `AX_columnMeasure` (Igusa box/shell single-coordinate Haar value, the per-column input — the closed form $(1 - Q^{-1})^V Q^{-A}$ is then *proved* in `L4.cellVolume_eq`); `omCells` (the finite GMN order- r cell partition; GMN §2.5 Def 2.21); `descend_size_lt` (strict cluster-size drop; GMN Cor 1.20/3.8, Lemma 3.11(3)).

3 The capstone ledger (#print axioms, verbatim)

The footprints below were captured with `lake env lean LeanUrat/AxChk.baseline.lean` against the built library (regenerated for this certificate; 55 declarations, 0 `sorryAx`). CORE abbreviates `{propext, Classical.choice, Quot.sound}`.

3.1 The two Goal theorems (the headline)

Declaration	Axiom footprint (verbatim)
<code>Goal.goal_theorem_montes</code>	<code>[propext, Classical.choice, tame_functionalEquation, Quot.sound]</code>
<code>Goal.goal_theorem</code> (measure route)	<code>[propext, Classical.choice, tame_functionalEquation, Quot.sound, PadicMeasure.AX_cellRecursion, PadicMeasure.clusterMeasure, PadicMeasure.descend_size_lt, PadicMeasure.omCells]</code>

Reading. The count-native capstone `goal_theorem_montes` carries *only* `tame_functionalEquation` on top of CORE: the entire measure layer (`clusterMeasure`, `AX_cellRecursion`, `omCells`, `descend_size_lt`) is *absent*, and there is no `sorryAx`. Its rationality flows through the pure-counting path (`MontesData.countingDensity`) its palindromy through (A3) transferred by the $\mathbb{Q}(t)$ identity theorem (L7). (A1), (A2) and the bridge (A4) are carried as *hypotheses* (M, D, F, `hbridge`), not axioms. The older `goal_theorem` retains the full measure route and so carries the measure axioms; both are kept for the historical derivation.

3.2 Rationality / decomposition (count-native, only-Montes) — all Core

Declaration	Footprint
<code>MontesData.coeff_isRational</code>	CORE
<code>MontesData.countingDensity_isRational</code>	CORE
<code>MontesData.countingDensity_eq_sum_coeff</code>	CORE
<code>MontesData.nodeMultiplicativity</code> (<i>derived</i> , was a field)	CORE
<code>MontesData.undecidedVanishes</code> ($U_N \rightarrow 0$, <i>derived</i>)	CORE
<code>OMInduction.clusterVol_isRational</code> (geometric collapse)	CORE
<code>RatFn.isRationalFn_listSum / _div / _qpow</code>	CORE
<code>CountingModel.density_le_decided_add_undecided</code>	CORE
<code>CountingModel.decided_le_density</code>	CORE
<code>CountingModel.countingDensity_is_squeezed_limit</code>	CORE

The decomposition $\rho = \sum_T C_T$, the rationality of each C_T , the generic OM-order induction, the squeeze semantics, and the rational-function closure lemmas all rest on CORE only — **no** measure axiom, no `tame_functionalEquation`, no `sorryAx`. This is the only-Montes footprint.

3.3 The proved building blocks — all Core

Declaration	Footprint
<code>L7.isPalindromic_of_agree</code> , <code>L7.tame_to_all_primes</code>	CORE

Declaration	Footprint
L7.ratfunc.agree_of_infinite, L7.eval.reflect_eq	CORE
L3.card_monicDegree, L3.irreducible_separable	CORE
L3Squarefree.card_squarefreeMonicDegree	CORE
L3Gauss.gauss_necklace_count,	CORE
L3Gauss.sum_e_mul_card_eq_pow	
L4.cellVolume_eq, L4.cellVolume_pIndependent	CORE
L5fix.selfLoop_geometric	CORE
M1.projPointCount_mul, M1.clusterLaw_baseChange	CORE
R1.treeDistanceWeight_values	<i>no axioms (decide)</i>
NestedCollapse.nested_reproduces_I2 / _I3	CORE
NestedCollapse.nested_block_lt	CORE
NestedCollapse.geometricLimit_of_selfLoop	CORE
MontesAxiom.RescaleC1.selfLoopExponent_eq_blockExponent	CORE, sub_one
Witness.montes_bundle_nonempty	CORE
Witness.montes_full_instance	CORE,
	tame_functionalEquation
Witness.DensityFoundation_nonempty	CORE,
	PadicMeasure.clusterMeasure

DensityFoundation_nonempty carries clusterMeasure because its witness references the old measure route; it is the OLD $\forall F$ goal witness and is **off** the only-Montes path. The complete coupled non-vacuity witness for goal_theorem_montes is montes_bundle_nonempty (CORE) / montes_full_instance (CORE, tame_functionalEquation).

3.4 The measure layer (PadicMeasure, L6M4, BB3inf) — where the wall lives

Declaration	Footprint
PadicMeasure.clusterMeasure_isRational_of_AX	CORE, AX_cellRecursion, clusterMeasure, descend_size_lt, omCells
PadicMeasure.clusterMeasureModel_rec	CORE, descend_size_lt, omCells
PadicMeasure.T_BB3, residualCountFn_isRational	CORE
PadicMeasure.cellCoeff_isRational, bb1_isRational, omPivot_isRational, natCard_Fr	CORE
PadicMeasure.descend_children_of_factors	CORE, omCells
PadicMeasure.cell_stratum_of_residual	CORE, omCells
L6M4.density_rational, L6M4.sum_clusterVol_rational	CORE, AX_cellRecursion, clusterMeasure, descend_size_lt, omCells
BB3inf.clusterVolume_rational_of_OM	CORE, BB3inf.descend, BB3inf.descend_size_lt
BB3inf.T_BB3, T_BB1, bb3CellCount_isRational	CORE

Declaration	Footprint
<code>Interface.clusterVolume_rational</code>	CORE, clusterMeasure, omCells

4 Which parts of the paper’s argument are formalized

The table maps the paper’s proof skeleton to Lean. *Proved* (CORE) means a Lean `theorem` resting only on CORE. *Assumed* means carried as an (A1)/(A2)/(A3) hypothesis or measure-route axiom.

Paper (§/label)	Lean theorem(s)	Status
Main theorem (<code>thm:main</code>)	<code>Goal.goal_theorem_montes</code>	Proved <i>modulo</i> (A1)(A2)(A3)
Tree decomposition $\rho = \sum_T C_T$ (<code>eq:tree-decomp</code>)	<code>MontesData.countingDensity_eq_sum_coeff</code> (genuine limit-interchange, <i>not</i> <code>rfl</code>)	Proved (CORE)
BB1 Newton-cell box volume $(1 - Q^{-1})^V Q^{-A}$ (<code>lem:bb1</code>)	<code>L4.cellVolume_eq</code> , <code>MontesAxiom.boxVolume_eq</code>	Proved (CORE)
BB3 residual count over $\mathbb{F}_q$, $= Q^{d_S-1}$ (<code>lem:bb3</code>)	<code>MontesAxiom.residualBoxCount</code> ($d_S \geq 2$), <code>PadicMeasure.T.BB3</code>	Proved (CORE)
Per-shape rationality by OM-order induction (<code>prop:per-shape</code>)	<code>OMInduction.clusterVol_isRational</code> (well-founded on <code>treeSize</code>)	Proved (CORE)
Geometric self-loop collapse (the wild absorption, <code>rem:wild-absorbed</code>)	<code>NestedCollapse.geometricLimit_of_selfProp</code> , <code>L5fix.selfLoop_geometric</code> , <code>R1.treeDistanceWeight_values</code>	Proved (CORE)
Box additivity / per-node multiplicativity	<code>MontesAxiom.clusterCount_boxSum</code> , <code>nodeMultiplicativity</code> (derived)	Proved (CORE)
$U_N \rightarrow 0$ (undecided pool vanishes)	<code>MontesData.undecidedVanishes</code> (squeeze)	Proved (CORE)
Monic $\rightarrow$ projective bridge (<code>sec:bridge</code>)	<code>M1.projPointCount_mul</code> , <code>clusterLaw_baseChange</code>	Proved (CORE)
Uniform rationality corollary (<code>cor:rationality</code>)	<code>MontesData.countingDensity_isRational</code> , <code>coeff_isRational</code>	Proved (CORE)
Palindromy at wild primes (<code>prop:palindromy</code>)	<code>L7.tame_to_all_primes</code> , <code>isPalindromic_of_agree</code> (via $\mathbb{Q}(t)$ identity theorem)	Proved (CORE), uses (A3)
<i>The measure inputs (assumed, not formalized as proofs):</i>		
p -adic integration / Haar normalization (<code>sec:padic-integration</code>)	(A1) <code>boxHaarEquidist</code> + <code>nodeMeasure_boxSum</code> ; measure route	Assumed
GMN OM-tree machinery (<code>hyp:gmn</code> , <code>sec:om</code>)	(A2) <code>cells</code> , <code>cells_descend</code> , <code>finiteTermination</code> ; measure route <code>omCells</code> , <code>descend_size_lt</code>	Assumed

The proof *architecture* — the tree decomposition, BB1, BB3, the OM-order induction with geo-

metric collapse, the $U_N \rightarrow 0$ squeeze, the monic $\rightarrow$ projective bridge, and the palindromy transfer — is formalized as CORE Lean theorems. The *measure inputs* (existence and normalization of the p -adic Haar volume; the GMN cell/descent structure) are assumed.

5 The measure wall (the honest limitation)

mathlib v4.31.0 provides no p -adic Haar measure (no `MeasureSpace/haarMeasure` on `PadicInt` or any DVR; a search of `Mathlib/MeasureTheory` for `Padic` is empty). Consequently:

- The paper’s actual density $\rho(n, \sigma; q)$ — a genuine p -adic Haar volume — **cannot be constructed** in Lean. The capstones quantify over an abstract `DensityFoundation/CountingModel`; they are *conditional* theorems.
- Non-vacuity *is* witnessed: the full coupled `goal_theorem_montes` bundle has a sorry-free inhabitant (`Witness.montes_bundle_nonempty / montes_full_instance`), with non-trivial density $g_\sigma \equiv 1$ and (A1) holding by `rfl (boxMeasure:=countCellCoeff)`. *But* the only inhabited instance is a trivial degree-0 witness ($q=2, n=0$: `trivM/trivD/trivF`), **not** the paper’s ρ . Identifying any constructed instance with ρ is exactly the un-crossed wall.
- Concretely, the wall is (A1): that the per-box p -adic Haar volume equals the separately *proved* lattice/finite-field value. The right-hand sides of these identities are proved; only their *identification with the Haar volume* is assumed.

We therefore describe the formalization as *formally verified modulo the explicit list of Section 2*, never as “formally verified” bare. The r -point/pole-control refinements and any conjectural higher-order item are **not** claimed as proven here.

6 Pre-existing flagged sorrys (off every capstone footprint)

Two declarations in `LeanUrat.L3` carry a flagged `sorry`: `L3.card_squarefreeMonicDegree` and `L3.gauss_necklace_count`. Both require generating-function / Möbius identities absent from mathlib v4.31.0. They are **re-proved sorry-free** as standalone theorems: `L3Squarefree.card_squarefreeMonicDegree` (the $A \cdot B^2$ unique-factorization count) and `L3Gauss.gauss_necklace_count` (the necklace identity plus Möbius inversion). Because `L3Squarefree` and `L3Gauss` both *import* `L3`, wiring the proved statements back into `L3` would create an import cycle; the cycle blocks the wiring, so the two `L3 sorrys` remain. This is benign: the count-native capstone path consumes the *sorry-free* versions (via `T_BB3 / residualBoxCount`), never the `L3` stubs. No capstone `#print axioms` footprint contains `sorryAx` (verified: 0 occurrences across all 55 audited declarations).

7 Independent cross-checks (EVIDENCE, not proof)

The computations below were run *outside* Lean and are *evidence* that the closed-form arithmetic the Lean modules prove symbolically matches independent computation; they are not part of the formal proof. Two reproducible scripts (system `python3 + sympy 1.13.3`):

- `uniform-rationality/_cert_crosschecks.py` — 25 exact checks (sympy identities + exhaustive $\text{GF}(Q)$ enumeration; no floats). All pass.
- `uniform-rationality/_reconcile_om_bcfg.py` — the symbolic OM density engine vs BCFG, $n = 2 \dots 6$; output `results/reconcile_om_bcfg.json` (`all_match: true`).

7.1 Closed-form arithmetic the Lean modules prove (exact)

Lean claim	Independent check	Result
R1.treeDistanceWeight, L5fix.selfLoopExponent: $w(s) = s(s+1)/2 - 1$	$w(2, 3, 4, 5) = 2, 5, 9, 14$ (exact)	match
C1 selfLoopExponent_eq.blockExponents_sub_one; 7 (exact $\mathbb{N}$) $w(s) = (s + \binom{s}{2}) - 1$		match
NestedCollapse.closedI2: $I_2 =$ $q/(q+1)$ via $a/(1-r)$, $a = (q-1)/q$, $r = q^{-2}$	$\text{sympy.simplify} \equiv 0$	match
NestedCollapse.closedI3: $I_3 =$ $\frac{q^3(q^2 - q + 1)}{(q+1)\Phi_5}$ from nestedRHS	$\text{sympy.simplify} \equiv 0$	match
L4.cellVolume_eq: column product = $(1 - Q^{-1})^V Q^{-A}$	$(V, A) \in \{(1, 1), (2, 3), (3, 5)\}$, exact	match
L3Squarefree.card_squarefreeMonicDegree(Q) enum, $Q \in \{2, 3\}$, $d = 2, 3$ ($d_S \geq 2$): $S(m) = Q^m - Q^{m-1}$		match
L3Squarefree.sum_identity (recursion (*)): $Q^n = \sum_{2j \leq n} S(n-2j)Q^j$	$Q \in \{2, 3, 5\}$, $n = 0, \dots, 8$ exact	match
L3Gauss: $d N_d = \sum_{e d} \mu(e)Q^{d/e}$, $\sum_{e d} e N_e = Q^d$	$\text{GF}(Q)$ enum, $Q \in \{2, 3, 5\}$, $d \leq 4$	match
M1.projPointCount_mul: $(q - 1) \sum_{i=0}^n q^i = q^{n+1} - 1$	$n = 0, \dots, 5$ exact	match
T_BB3/residualBoxCount ($d_S \geq 2$): $\#\{\text{non-squarefree monic deg } d\} = Q^{d-1}$	$\text{GF}(Q)$ enum, $Q \in \{2, 3\}$, $d = 2, 3$	match

Honest note on hypotheses. The closed forms $S(m) = Q^m - Q^{m-1}$ and $\#\{\text{non-squarefree}\} = Q^{d-1}$ are the *Lean theorem statements*, which carry the hypothesis $2 \leq m$ (resp. $2 \leq d_S$): at $m = 1$ all Q degree-1 monics are squarefree, so $S(1) = Q \neq Q - 1$ and the non-squarefree count is $0 \neq Q^0$. The cross-check mirrors that hypothesis exactly (it tests $d \geq 2$). The recursion (*) holds for all n (using the correct $S(0) = 1$, $S(1) = Q$, $S(m \geq 2) = Q^m - Q^{m-1}$). All $\text{GF}(Q)$ rows are genuine exhaustive enumerations of monic polynomials over a prime field, so they verify the finite-field counts directly. No floating-point step.

7.2 The symbolic OM engine reproduces BCFG root-count moments through $n = 6$

The OM density engine (`uniform-rationality/om.density_engine.py`) assembles the *full per-type* symbolic distribution $\rho(n, \sigma; q)$ (e.g. all 34 types at $n = 6$, including the first deep-wild OM-order-3 branch), aggregates to the root-count distribution $P(n, r; q)$, and compares to BCFG's independently-derived $P(n, r; q)$ (arXiv:2101.09590 Thm 1; generating-function inversion, methodologically independent of the OM/cluster engine). Every comparison is an *exact* rational-function identity `sympy.simplify(OM-BCFG) = 0`, re-run for this certificate (`results/reconcile_om_bcfg.json`, `all_match: true`).

n	#types	support r	all $P(n, r; q)$ match BCFG
2	3	$\{0, 2\}$	yes (3 cells)
3	5	$\{0, 1, 3\}$	yes (4 cells)
4	11	$\{0, 1, 2, 4\}$	yes (5 cells)
5	17	$\{0, 1, 2, 3, 5\}$	yes (6 cells)
6	34	$\{0, 1, 2, 3, 4, 6\}$	yes (7 cells)

Additional exact (`simplify(diff) = 0`) checks in the same run: $\sum_r P(n, r; q) = 1$ at every n ; $\sum_\sigma \rho(n, \sigma; q) = 1$ (the exhaustive enumeration gate); $E[N] = 1$ for all n ; the stabilized moments $E[N^2] = (2q^4 + q^3 + 3q^2 + q + 2)/\Phi$ and $E[(N)_2] = (q^2 + 1)^2/\Phi$ for $n \geq 3$; and the factorial moments $E[(N)_d]$ matching BCFG's d -th-moment stabilization ($n \geq 2d - 1$). The moments are *also* derivable via the Vandermonde/Selberg local-integral route (Track A), matching BCFG through $d \leq 4$ (`trackA_highd_moments.py`).

Per-type oracle (closes the within- r caveat). $P(n, r; q)$ is a per- r aggregate; two engines could agree on every $P(n, r)$ while disagreeing on individual type-densities. A PARI `factorpadic` projective oracle probed the *individual* densities $\rho(6, \sigma; p)$ at $p \in \{2, 3, 5, 7\}$, $N \sim 30\text{k}/\text{prime}$: 0 observed-but-unpredicted types, overall $\max |z| = 3.66$ (sign-flipping across primes — MC Poisson noise, not a systematic error), and every deep-wild order-3 frontier type matches within noise. This per-type agreement is empirical (MC), not a proof.

7.3 $n \leq 3$ exact closed forms vs BCFG, and the wild atoms

Claim (paper / engine)	Value, vs BCFG	Result
$\rho(3, \{(1, 1)^3\}; q)$ (all rational roots)	$\frac{(q^2 + 1)^2}{6\Phi}$	match
$P(3, 0; q)$ (no rational root)	$\frac{(q^2 + 1)^2}{3\Phi}$	match
$P(3, 1; q)$	$\frac{q^4 + 2q^3 + 2q + 1}{2\Phi}$	match
$n = 3$ five type densities (paper Ex.)	all palindromic, $\sum = 1$	match
First wild atom $n = 4$, slope- $\frac{1}{2}$, conditional law	$P((4, 1) \mid \text{insep.}) = \frac{q}{q + 1}$	match
vs recorded oracle (incl. wild $p = 2$ Artin-Schreier)	obs 0.6664 at $p=2$ vs $q/(q+1) = 0.6667$ ($N=80\text{k}$)	match
oracle at $p = 3, 5, 7, 11, 13$	0.7515, 0.8329, 0.8756, 0.9147, 0.9289	within 4σ
Deep-wild $n = 6$ order-3 cluster law $\beta_{(6,1)}(q)$	$\frac{q^2(q-1)(q^2+1)}{q^5-1}$, $\beta_{(6,1)}(2) = \frac{20}{31}$	match
rejects naive guess $q/(q+1) = \frac{2}{3}$	oracle $\approx 11\sigma$ rejection	reject

The $n = 4$ wild $p = 2$ row is the deepest evidence: the char-dependent geometry (a punctured parabola $\bar{b}^2 = 4\bar{a}$ at odd p versus a punctured line $\bar{b} = 0$, $R = (y + \sqrt{\bar{a}})^2$, inseparable at $p = 2$) yields the *same* conditional law $q/(q+1)$. The recorded oracle (`results/n4wild_insep.p{2,3,5,7,11,13}_s30*.json`, $N = 80\text{k}/\text{prime}$ aggregated) matches the exact prediction within MC noise at every prime, including wild $p = 2$ at deviation 0.0003 (4σ band 0.0067). The PARI `factorpadic` binary is unavailable

in this build environment; these rows reuse the previously recorded oracle data as the empirical anchor and re-verify the symbolic predictions and the comparison, which are reproducible.

8 Release module list

The library `LeanUrat` (target of `lake build LeanUrat`) comprises: `Interface`, `PadicMeasure`, `OMInduction`, `RatFn`, `CountingModel`, `MontesAxiom`, `Decomposition`, `NestedCollapse`, `L3`, `L3Squarefree`, `L3Gauss`, `L4`, `L5fix`, `L6M4`, `L7`, `M1`, `R1`, `BB3inf`, `Goal`, `Witness`, `Smoke` (+ `Basic`, the lake template placeholder; `AxChk_baseline`, the audit harness that regenerates Section 3).

Provenance statement

The axiom footprints in Section 3 are reproduced verbatim from `#print axioms` run on the built library via `lake env lean LeanUrat/AxChk_baseline.lean` (55 declarations, 0 `sorryAx`). The cross-checks in Section 7 are exact symbolic / exhaustive finite-field computations (`_cert_crosschecks.py`, 25/25 pass) and the OM–BCFG reconciliation (`_reconcile_om_bcfg.py`, `all_match: true`), run independently of Lean. We have stated the measure wall and the two flagged L3 `sorry`s plainly; no conjectural item (r -point / pole control) is presented as proven.